

An Evaluation Framework for Textile Stretch Sensor Layouts Based on Mutual Information

Jiaqi Mo
College of Design and Innovation,
Tongji University
Shanghai, China
mojiaqi@tongji.edu.cn

Nianchong Qu
College of Design and Innovation,
Tongji University
Shanghai, China
qnc@tongji.edu.cn

Yiran Wan
College of Design and Innovation,
Tongji University
Shanghai, China
2251992@tongji.edu.cn

Leheng Chen
College of Design and Innovation,
Tongji University
Shanghai, China
lokhang@tongji.edu.cn

Yu Xiao
Department of Information and
Communications Engineering,
Aalto University
Espoo, Finland
yu.xiao@aalto.fi

Qi Wang*
College of Design and Innovation,
Tongji University
Shanghai, China
qiwangdesign@tongji.edu.cn

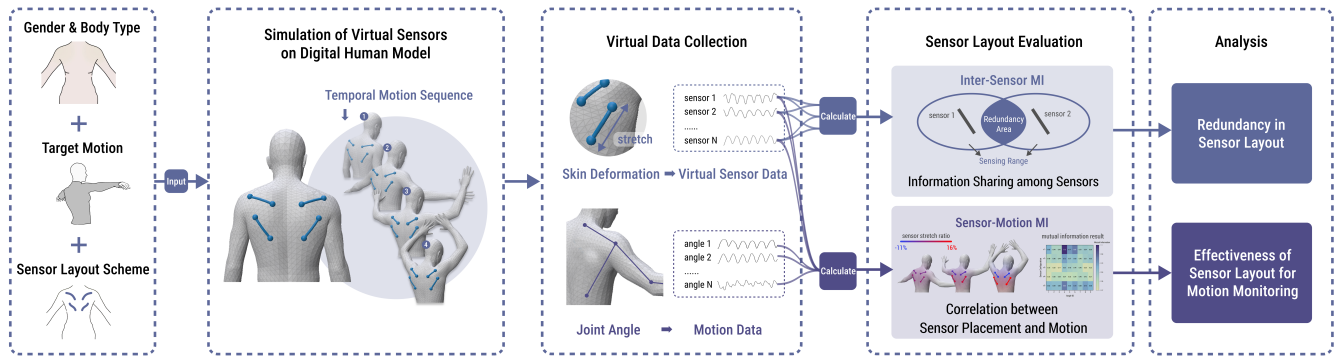


Figure 1: Evaluation framework for textile stretch sensor layouts. The framework takes user characteristics and target motions as inputs, constructs digital human model with virtual sensor deployment, and generates sensor data and motion data. Mutual information analysis quantifies both sensor-motion correlation and inter-sensor redundancy to assess layout effectiveness for motion tracking. The scalable framework can accommodate different sensor types and motion monitoring tasks.

Abstract

Textile stretch sensors demonstrate significant potential in motion capture. However, sensor layouts critically influence motion monitoring performance, necessitating effective evaluation of sensor placement strategies. Current methods lack systematic approaches for rapid assessment of multiple sensor layouts. This paper presents an evaluation framework for textile sensor layouts. The framework employs mutual information to quantify correlations between sensor positions and joint angles, as well as inter-sensor relationships,

based on simulated data, evaluating the effectiveness of sensor layouts under specific motion patterns. This approach enables low-cost, rapid assessment without physical prototypes. A preliminary case study demonstrates its feasibility and potential. This framework provides a computational foundation for textile sensor deployment optimization.

CCS Concepts

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

Keywords

Textile sensors, Wearable Motion Capture, Sensor Layout Evaluation, Mutual Information, Virtual Sensors

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*Corresponding author.

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1 Introduction

Textile-based stretch sensors have attracted considerable attention in human motion capture applications due to their flexibility and comfort [6], demonstrating significant potential in sports [28], rehabilitation [29], and interactive system [15]. Among various types of stretch sensors, knitted metal fiber sensors are frequently applied in wearable motion monitoring systems [13]. When integrated into tight-fitting garments, these strain sensors respond to body movements that induce fabric deformation, altering the contact area between metal fibers and resulting in measurable electrical signals for continuous and non-intrusive motion monitoring [26]. The effectiveness of sensing systems critically depends on the strategic sensor placements across the human body, as different locations provide varying motion information quality [12].

Existing approaches for evaluating sensor layouts still rely heavily on empirical, trial-and-error methods [31], and lack systematic frameworks for efficient assessment and optimization of multi-sensor arrangements. Although some studies have focused on analyzing data features from individual sensors [16], they often overlook multi-sensor coordination and spatial synergy, which are crucial for holistic performance evaluation. Others employ machine learning and feature elimination techniques to guide sensor selection [3], yet these approaches are typically data-dependent and require extensive physical experimentation and repeated prototype fabrication, leading to high iterative costs. Moreover, such methods are susceptible to variations across different models and datasets, limiting their generalizability.

To address these challenges, we propose a simulation-based evaluation framework to assess stretch sensor layouts in tight-fitting garments. Our approach leverages mutual information, an information-theoretic measure that quantifies the statistical dependency between two variables, to characterize both the relationships between sensor positions and motion patterns, and the information sharing among multiple sensors [20]. By simulating sensors with digital human models, the framework enables rapid layout assessment and analysis without the need for physical prototyping. A preliminary case study demonstrates the feasibility of our evaluation approach by applying the framework to assess existing sensor layouts for shoulder abduction angle estimation [32].

The main contribution of this work is a systematic evaluation framework for textile stretch sensor layouts. Mutual information is introduced to quantify the correlations between sensor positions and joint angles, as well as inter-sensor relationships based on the simulated human body and sensor data. This framework provides a computational approach for comparing layout schemes, enabling more efficient and systematic sensor placement decisions.

2 Related Work

Textile sensors are well-suited for long-term daily motion monitoring [9] and have been applied to joint angle prediction [23], posture estimation [34], and motion pattern classification [1]. However, sensor layouts significantly impact motion capture accuracy [31], making sensor placement critical for sensing system design.

Textile sensor placement strategies have evolved from experience-driven to technology-assisted approaches, including anatomy analysis [17], photographic analysis [21], and optical motion capture system [8]. Simulation-based approaches have emerged. Imamura et al. utilized digital human models and virtual bend sensors to optimize sensor layouts for joint angle estimation [10].

Although these studies propose various layout strategies, evaluating sensor layout effectiveness remains challenging. Current evaluation includes qualitative and quantitative methods. Qualitative analysis relies on anatomical and kinematic principles [24], providing theoretical guidance but lacking objective quantification. Quantitative evaluation encompasses two main approaches. Signal statistical analysis methods evaluate the correlation between sensor and motion through correlation coefficients [16], fuzzy entropy [4], and PCA contribution analysis [24]. Machine learning-based methods employ recursive feature elimination [11], ablation experiments [18], or indirect evaluation through downstream task performance [3, 25]. However, statistical methods struggle with multi-sensor complexity, while machine learning approaches lack objectivity due to dependence on parameter tuning and model selection. In contrast, mutual information demonstrates strength in analyzing complex relationships among multiple variables [33], providing a viable approach for multi-sensor layout evaluation.

These quantitative methods requires substantial data support, facing challenges of high cost and time-consuming data collection. To improve evaluation efficiency, Li et al. developed a plug-and-play e-textile platform for repeatable performance testing and rapid sensor placement exploration [14]. Researchers have also begun exploring simulation approaches to address these issue. Park et al. simulated the interaction between flexible sensor and human motion using musculoskeletal simulation [22]. Wang et al. proposed a simulation-based approach to optimize the wiring layout for minimizing motion resistance [27]. The Skinned Multi-Person Linear (SMPL) parametric human body model has proven effective for textile sensor simulation by representing human body shape and pose through statistical parameters [19]. Fang et al. utilized SMPL models for elbow joint textile sensor modeling [7]. These methods provide data-driven insights for sensor layout evaluation.

To address evaluation challenges, we propose an evaluation framework that utilizes mutual information and simulation for sensor placement analysis and decision-making.

3 Evaluation Framework

We propose a systematic evaluation framework for textile sensor layout, as shown in Figure 1. The framework employs mutual information as a core evaluation metric to analyze sensor layout effectiveness and redundancy. A simulation-based process is developed using digital human models and virtual sensors to generate sensor data and motion data for layout evaluation through mutual information. This framework enables rapid assessment of textile sensor layouts without physical prototypes.

3.1 Simulation of Virtual Textile Sensor on Digital Human Body

The simulation process provides the computational and analytical data foundation for sensor layout evaluation.

Digital human models are constructed using the SMPL model [19], parameterized based on user gender and body type, and motion sequences driven by target motion.

Virtual textile sensors are modeled by defining attachment points as SMPL vertices on the skin surface, with sensor length represented as the geodesic distance between these points, as shown in Figure 2. According to the target sensor position, length and orientation parameters, the corresponding vertices are identified as attachment points in the initial digital human body state. Sensor strain is calculated by tracking the 3D coordinates of attachment points over time and computing geodesic distance variations using the heat method [5]. The resulting strain values can be further converted to sensor signals through established strain-signal characteristics [7].

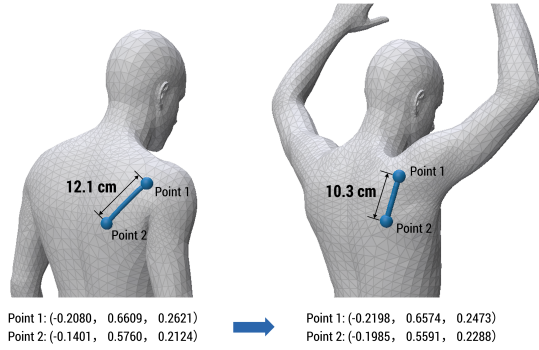


Figure 2: Virtual textile sensor placed on the digital human model.

Motion information is extracted from the digital human model by selecting vertices corresponding to bony landmarks as key tracking points. Following the International Society of Biomechanics (ISB) definition [30], these points establish standardized Joint Coordinate Systems (JCS) for calculating target joint angles.

3.2 Mutual Information-Based Sensor Layout Evaluation

Based on the virtual sensor data and motion data generated through simulation, the framework evaluates sensor layout effectiveness by quantifying the extent to which sensors capture target motion information. We adopt mutual information (MI) as the core metric, establishing a comprehensive evaluation through two key dimensions: the sensor-motion correlation and inter-sensor information sharing. Single-sensor analysis forms the foundation of layout evaluation, as sensor response is influenced by its placement, orientation, and movement amplitude, directly reflecting its inherent sensing capability. For complex motion monitoring scenarios, the deployment of multiple sensors is typically required, necessitating the assessment of their collective representational capacity and identification of potential information redundancy within the sensor network.

MI provides an intuitive and rigorous measure of how much knowing one variable reduces the uncertainty of another, thus capturing the shared information between sensor signals and motion variables. Compared to traditional linear correlation methods, MI offers several key advantages: (1) It is capable of capturing complex,

non-linear dependencies that are common in motion and strain data but often overlooked by linear metrics. (2) MI imposes no assumptions on data distribution, making it suitable for diverse and non-Gaussian signal characteristics frequently encountered in wearable sensing systems [20]. (3) It can be extended to multi-variate analysis, enabling a more holistic evaluation of inter-sensor information relationships within a networked layout.

For mutual information calculation, we use a k -nearest neighbor (k -NN) approach with the Chebyshev distance metric. This non-parametric method is robust to high-dimensional, nonlinear relationships between sensor inputs and motion outputs. For each sample point, the distance to its k^{th} nearest neighbor in the joint space (sensor + angle) is computed, and the number of neighbors within this radius in the marginal spaces is counted. The mutual information is estimated using the following KSG-based formula:

$$MI \approx \psi(k) + \psi(n) - \langle \psi(n_x + 1) + \psi(n_y + 1) \rangle \quad (1)$$

where $\psi(\cdot)$ is the digamma function, n is the number of samples, and n_x , n_y are the number of neighbors in the sensor and angle marginal spaces, respectively.

In real-world scenarios, skin deformation measurement is challenging and typically requires indirect inference from sensor outputs, introducing potential noise and mapping errors. Our virtual sensor framework bypasses this limitation by directly accessing strain values during simulated body motion, eliminating the need for signal-to-deformation conversion. This direct strain acquisition preserves the essential sensor response characteristics and provides a more accurate foundation for mutual information analysis, ultimately enabling more reliable evaluation and optimization of sensor placement in wearable systems.

4 Preliminary Case Study

To preliminarily validate this evaluation framework, we conducted a case study on shoulder joint monitoring by reproducing the sensor layout proposed by Zhang et al. [32], which proved effective in upper limb motion capture. As reported in Zhang et al.'s study, five sensors were placed around the anterior and posterior fascicles of the deltoid, trapezius, pectoralis major, and levator scapulae muscles to monitor arm abduction, extension, and flexion.

We selected a male model from the D-Faust dataset [2] with "chicken wings" movement, characterized by shoulder joint abduction and adduction around the sagittal axis with a range from 50° to 120°. Virtual sensors were positioned on the digital human model as illustrated in Figure 3, with sensor strain calculated as geodesic distance variations between attachment points. Motion data was extracted following ISB standards using 14 bony landmarks on model vertices to establish shoulder joint coordinate system. Coordinate systems for the trunk, scapula, and humerus were established to calculate 9 Euler angles based on their relative orientations.

During shoulder movement simulation, we simultaneously collected virtual sensor strain data and corresponding shoulder joint angle data. Mutual information was calculated between sensors and joint angles to quantify sensor-motion associations, and between sensor pairs to assess information redundancy within the network.

The results are shown in Figure 4 and Figure 5, where sensor configurations are denoted as S(1, 2, 3, 4, 5) representing the combination of sensors 1 through 5. In Figure 4 (a), the results reveals

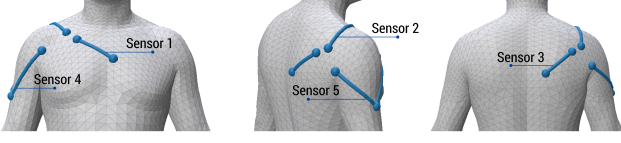


Figure 3: Reproduction of literature-based five-sensor layout using virtual sensors on digital human model.

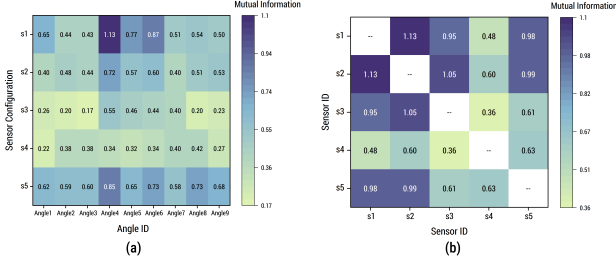


Figure 4: Heatmaps of mutual information results for shoulder movement: (a) Mutual information between single-sensor configurations and nine Euler angles; (b) Mutual information between sensors.

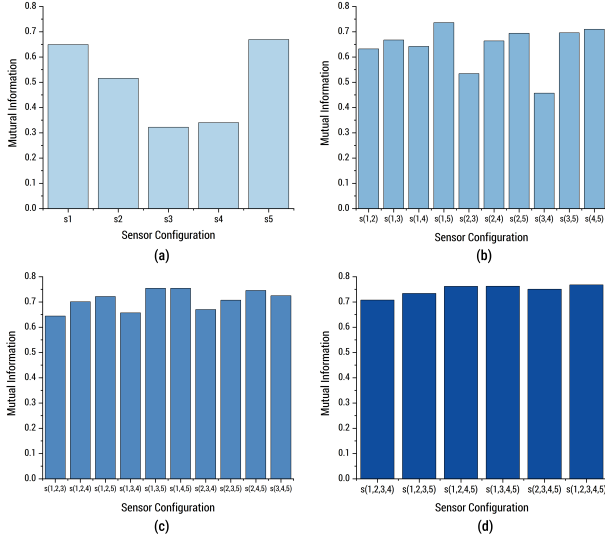


Figure 5: Average mutual information between sensor configurations and nine Euler angles. (a)-(c) represent single-, dual-, and triple-sensor configurations respectively; (d) represents four- and five-sensor configurations.

distinct performance characteristics among sensors. Sensors 1 and 5 consistently exhibit high MI values with joint angles, indicating that their placement positions are highly correlated with the motion patterns. In contrast, Sensors 3 and 4 demonstrate lower MI values, suggesting limited individual contribution to motion information and potentially suboptimal placement positions.

Figure 4 (b) shows inter-sensor redundancy analysis, revealing efficiency concerns in the current layout. High mutual information between Sensors 1 and 2 (1.13) and between Sensors 1 and 5 (0.98) indicates substantial information overlap, potentially reducing system efficiency. Conversely, Sensor 4 demonstrates low inter-sensor mutual information (0.36-0.63) despite poor individual performance, suggesting it provides unique complementary information.

Figure 5 reveals that sensor-motion information capture exhibits diminishing returns as sensor count increases. While moving from single to dual-sensor configurations yields significant improvements, gains diminish notably beyond three sensors. Some three-sensor configurations (e.g., S(1, 3, 5)) achieve information capture comparable to certain four-sensor setups, suggesting that three to four sensors may provide near-optimal performance [4].

Based on these results, we could propose layout optimization strategies. The optimal three-sensor configuration S(1, 4, 5) achieves the highest average mutual information (0.75), while the four-sensor configuration S(1, 3, 4, 5) approaches five-sensor performance with reduced redundancy. Both configurations strategically combine high-performing sensors (1 and 5) with complementary sensor (4), balancing representation capability and information diversity.

5 Conclusion and Future Work

In this study, we proposed an evaluation framework for textile strain sensor layouts in wearable motion monitoring systems. The framework utilizes mutual information and virtual sensors to assess layout effectiveness while reducing dependence on physical prototyping, providing computational foundation for optimization. The preliminary case study demonstrates its potential for systematic multi-sensor layout assessment.

The current implementation is based on simplified assumptions about skin-garment interaction, and due to environmental and individual variability, applying simulation to real-world scenarios remains challenging. Future work will conduct comparative experiments in simulated and real environments to verify the simulation approach's reliability and effectiveness. To support personalized sensor layout evaluation, we plan to develop customizable parametric human models for flexible configuration of body shape, demographics, and motion sequences. Interactive design interfaces will be developed to facilitate rapid sensor layout assessment and iterative optimization.

Furthermore, we aim to generate extensive datasets through simulation for layout validation and model training in real-world fabric sensing applications. To achieve this, the strain-to-signal mapping relationship must be established to develop robust and generalizable sensor response models, while comprehensive studies on skin-garment deformation coupling are needed to accurately represent realistic sensor wearing scenarios.

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